

Dynamic-Envelope Re-Tuning of a Graph-Theoretic Swarm Formation Planner for School-Like Robotic-Fish Behavior: A Controlled Simulation Study

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Abstract—Distributed trajectory optimizers can fly a rigid swarm formation through cluttered space, but the motion they produce is shaped as much by the *dynamic-feasibility envelope*—the velocity and acceleration limits fed to the optimizer—as by the planner itself. Biological schools of fish move slowly and smoothly, with drag-limited accelerations, in marked contrast to the aggressive, jerk-rich trajectories typical of quadrotor swarms. We ask a focused, reproducible question: holding a state-of-the-art graph-theoretic formation planner and the obstacle field fixed, how does contracting the dynamic envelope from quadrotor values to fish-like values change the swarm’s kinematics, timing, formation accuracy, and safety? We re-skin the agents of an open-source seven-robot formation planner as a school of robotic fish and re-tune only their motion envelope ($v_{\max}: 1.5 \rightarrow 1.0$ m/s, $a_{\max}: 8.0 \rightarrow 2.5$ m/s²), then run a controlled headless study across five obstacle seeds per condition. The fish envelope roughly halves root-mean-square (RMS) jerk ($0.66 \rightarrow 0.34$ m/s³) and cuts mean acceleration by $\sim 44\%$, producing visibly smoother, lower-effort motion, at the cost of $\sim 44\%$ longer traversal time. Crucially, formation accuracy, inter-agent spacing, obstacle clearance, and a 100% mission-success rate are all preserved within run-to-run noise. The results quantify a clean time-versus-smoothness trade-off and show that a graph-theoretic formation planner degrades gracefully under a strongly contracted, bio-inspired dynamic envelope.

Index Terms—Swarm robotics, formation control, trajectory optimization, bio-inspired robotics, robotic fish, multi-robot systems.

I. INTRODUCTION

COORDINATED motion of a multi-robot team in a rigid, shape-preserving formation is a core capability for aerial inspection, collective transport, cooperative mapping, and underwater monitoring. The hard version of the problem couples three requirements that pull against one another: each robot must (i) hold its place in a desired geometric *formation*, (ii) avoid static obstacles and its neighbors, and (iii) respect its own *dynamic limits*. Recent distributed spatio-temporal optimizers solve all three jointly and in real time. In particular, the *Swarm-Formation* framework of Quan *et al.* [1], [2] introduces a differentiable, graph-theoretic measure of formation similarity and folds it into a per-agent trajectory optimizer built on the EGO-Swarm [3] and MINCO [4] machinery, enabling

a swarm of quadrotors to weave through a dense forest while holding shape.

A property that is easy to overlook is that the *shape* of the executed motion—how fast, how aggressively, how smoothly the swarm moves—is governed not only by the planner but by the *dynamic-feasibility envelope*, i.e. the maximum velocity $v_{\max}$ and acceleration $a_{\max}$ the optimizer is told to respect. Quadrotors are typically given a large envelope and consequently fly fast, jerky trajectories. Biological systems at the other extreme—a school of fish [5], a flock of birds [6]—move comparatively slowly and smoothly, their accelerations bounded by hydrodynamic or aerodynamic drag. This contrast motivates a concrete question with direct engineering relevance for slow, drag-limited platforms such as underwater vehicles [7], [8]:

If we keep a state-of-the-art formation planner and the environment fixed and only contract the dynamic envelope from quadrotor values to fish-like values, what exactly do we gain and what do we lose?

This paper answers that question with a controlled simulation study. We take the open-source *Swarm-Formation* planner unchanged, re-skin its seven agents as a school of robotic fish (a new mesh and color, plus the slower envelope), and run a paired, headless experiment over five obstacle seeds in each condition. Because everything but the envelope is held fixed, the resulting differences are attributable to the envelope alone, isolating its effect from confounds of planner or map.

Contributions. This paper makes three contributions:

- 1) A *bio-inspired re-instantiation* of a graph-theoretic quadrotor formation planner as a robotic-fish school, changing only the visual embodiment and the dynamic-feasibility envelope while leaving the optimization core untouched (Section III).
- 2) A *controlled, reproducible benchmark* (seven agents, hexagon template, identical random-forest seeds, headless 100 Hz logging) that isolates the envelope’s effect and reports kinematic, formation, and safety metrics with run-to-run statistics (Section IV).
- 3) A *quantified time-versus-smoothness trade-off*: the fish envelope halves RMS jerk and cuts mean acceleration $\sim 44\%$ at the cost of $\sim 44\%$ longer traversal, while formation accuracy, spacing, clearance, and a 100% success rate are preserved within noise (Section V).

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This work builds on the open-source ZJU-FAST-Lab *Swarm-Formation* framework [1].

We are explicit about scope: this is a re-skin plus envelope re-tune. The underlying control remains the second-order (SO(3)) quadrotor model with reduced limits rather than a full hydrodynamic, non-holonomic swimmer; the observed differences therefore reflect the dynamic envelope, not body-caudal-fin hydrodynamics [9]. Section VI discusses how the higher-fidelity model would extend the framework.

II. RELATED WORK

A. Formation and Flocking Control

Multi-agent formation control has a long history spanning leader-follower, virtual-structure, behavior-based, and rigidity-graph formulations [10]. Behavioral flocking traces to Reynolds’ boids [6] and has been realized on physical robot swarms [11]. These methods coordinate the team but do not, in general, produce smooth, dynamically feasible, obstacle-aware trajectories in dense clutter, which is the regime we study.

B. Optimization-Based Swarm Planning

A complementary line treats each robot’s path as a continuous trajectory and optimizes it against obstacle, smoothness, and dynamic-feasibility costs. EGO-Planner and EGO-Swarm [3] provide a gradient-based, ESDF-free local planner that scales to decentralized teams; MINCO [4] supplies a minimum-control polynomial parameterization with analytic spatial-temporal gradients. Building on these, Swarm-Formation [1] adds a differentiable graph-similarity term so that the swarm holds a desired shape while flying through a forest, later extended to a complete, robust navigation system [2]. We use this planner as our fixed substrate and vary only its dynamic envelope.

C. Bio-Inspired and Underwater Collectives

Fish schools coordinate through local interactions inferred from trajectory data [5], and robotic realizations such as Blueswarm demonstrate implicit 3D collective behaviors underwater using only on-board vision [7]. Such platforms are inherently slow and drag-limited [8], [9], which is precisely the dynamic regime our re-tuned envelope emulates. Our study does not propose a new swimmer; instead it asks how an existing high-performance formation optimizer behaves once pushed into this slow, smooth regime.

III. METHOD

A. Background: Graph-Theoretic Formation Cost

The planner represents the instantaneous swarm as a complete weighted graph whose N vertices are the agent positions $p_i \in \mathbb{R}^3$. Edge weights are squared Euclidean distances, $A_{ij} = \|p_i - p_j\|_2^2$, the degree is $d_i = \sum_j A_{ij}$, and the symmetric normalized Laplacian is

$$\hat{L}_{ij} = \begin{cases} 1, & i = j, \\ -A_{ij} d_i^{-1/2} d_j^{-1/2}, & i \neq j. \end{cases} \quad (1)$$

TABLE I
THE TWO EXPERIMENTAL CONDITIONS. ONLY THE DYNAMIC-FEASIBILITY ENVELOPE DIFFERS.

	Baseline	Robotic fish
$v_{\max}$ (m/s)	1.5	1.0
$a_{\max}$ (m/s ²)	8.0	2.5
Planner	Swarm-Formation (unchanged)	
Agents / template	7 / hexagon	
Start $\rightarrow$ goal	$x = -26 \rightarrow +26$ m	
Environment	random_forest, seeds 1–5	
Cost weights	identical	

Given a desired formation template with Laplacian $\hat{L}^{\text{des}}$, the *formation-similarity cost* is the squared Frobenius norm of the Laplacian difference,

$$f_{\text{form}} = \|\hat{L} - \hat{L}^{\text{des}}\|_F^2, \quad (2)$$

which is invariant to translation, rotation, and uniform scaling of the formation and is differentiable in the agent positions; its analytic gradient $\partial f_{\text{form}} / \partial p_i$ is propagated to each agent’s trajectory [1]. Each agent optimizes a MINCO [4] polynomial trajectory against a joint cost

$$J = \lambda_f f_{\text{form}} + \lambda_o f_{\text{obs}} + \lambda_s f_{\text{smooth}} + \lambda_d f_{\text{dyn}}, \quad (3)$$

where f_{obs} penalizes obstacle and inter-agent collision, f_{smooth} is the minimum-jerk control effort, and f_{dyn} penalizes violation of the dynamic-feasibility envelope $\{v_{\max}, a_{\max}\}$. **We do not modify** (2)–(3); they define the fixed substrate.

B. Robotic-Fish Re-Instantiation

Our intervention is deliberately minimal and surgical, so that any measured effect is unambiguously attributable to the dynamic envelope:

- **Embodiment.** Each agent’s render mesh is replaced by a stylized, laterally compressed fish body (caudal, dorsal, and pectoral fins, nose along $+x$, ≈ 1.2 units long) and colored fish-orange. This is purely visual and does not enter the optimizer.
- **Dynamic envelope.** The two feasibility limits are contracted from quadrotor to fish-like values: $v_{\max} : 1.5 \rightarrow 1.0$ m/s and $a_{\max} : 8.0 \rightarrow 2.5$ m/s². The lower acceleration ceiling emulates the gentle, drag-limited thrust of body-caudal-fin swimming [9].

Table I summarizes the two conditions. Everything except the two envelope parameters—planner, cost weights, formation template, number of agents, start/goal, and obstacle field—is held identical.

IV. EXPERIMENTAL SETUP

A. Task and Environment

Seven agents start in a hexagonal template (one centroid, six vertices) at $x = -26$ m and must traverse a random_forest obstacle field to a goal at $x = +26$ m in autonomous global-waypoint mode, holding the hexagon throughout. The forest is pseudo-randomly generated; we use five seeds (1–5), and *the same seed produces the same forest in both conditions*, making the comparison paired.

B. Protocol and Logging

Each run executes headless under ROS [12]. After a 6 s warm-up to let the planner initialize and the formation settle, we record each agent’s odometry and commanded control at 100 Hz, together with the obstacle map, for 90 s. The recording window is chosen to outlast the slowest condition: the fish need ≈ 64 s to traverse, so a shorter 60 s window (used in an earlier pilot) cut every fish run short of the goal and left traversal time undefined; 90 s lets both conditions complete with margin. With two conditions $\times$ five seeds, the study comprises ten logged runs.

C. Metrics

From the logged trajectories we compute, per run: *kinematics*—maximum and mean speed, maximum and mean acceleration, and RMS jerk (mean speed/acceleration are taken over *moving* samples, $\|v\| > 0.05$ m/s, so the stationary post-arrival tail does not bias them); *efficiency*—path length and traversal time (time for all seven agents to reach the goal); *formation quality*—the RMSE of pairwise inter-agent distances against the hexagon template, reported as the time-averaged mean and the worst-case maximum; and *safety*—minimum inter-agent distance, mean obstacle clearance, the fraction of samples within 0.3 m of an obstacle (near-miss fraction), and the binary mission-success flag (all seven agents arrive). We report mean $\pm$ standard deviation across the five seeds.

V. RESULTS

Table II reports the aggregate results; Figs. 1–4 visualize them.

A. Safety and Success Are Preserved

Both swarms succeed on every run: all seven agents reach the goal in all five seeds, in both conditions, with no inter-agent collisions (minimum separation ≈ 1.6 – 1.7 m, comfortably above the agent radius). Mean obstacle clearance (1.12 vs 1.09 m), near-miss fraction (0.058 vs 0.066), and worst-case formation RMSE (0.774 vs 0.799 m) all coincide within one standard deviation. The contracted envelope does not compromise safety.

B. Motion Becomes Slower and Much Smoother

The kinematic block of Table II shows the dominant effect. Maximum and mean speed fall $\sim 29\%$ and $\sim 26\%$, tracking the lower $v_{\max}$. The headline result is smoothness: *RMS jerk roughly halves* ($0.658 \rightarrow 0.338$ m/s³, -49%) and mean acceleration drops $\sim 44\%$. Fig. 2 shows the commanded speed and acceleration over the acceleration-rich opening phase: the fish profiles sit visibly lower and flatter, hugging their tighter dashed limits, whereas the baseline spikes toward its larger ceiling. This is exactly the gentle, low-effort motion one expects of a drag-limited swimmer, and it emerges purely from the envelope.

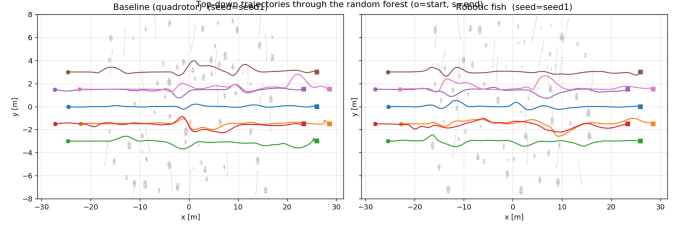


Fig. 1. Top-down executed paths through the random forest (seed 1), baseline (left) versus robotic fish (right). $\circ$ marks the start hexagon, $\square$ the goal. Both conditions traverse essentially the same corridors; path length differs by under 2%.

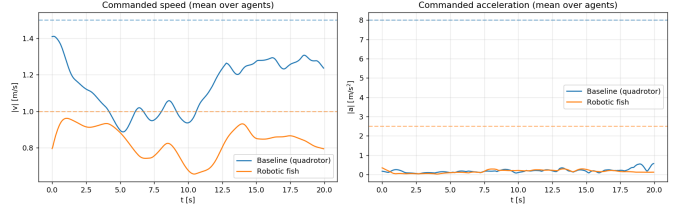


Fig. 2. Commanded speed (top) and acceleration (bottom), averaged over the seven agents, for the first 20 s. Dashed lines are the per-condition $v_{\max}$ and $a_{\max}$. The fish envelope produces lower, flatter, smoother profiles.

C. The Cost Is Time, Not Quality

Because the paths are nearly identical in length (51.4 vs 52.3 m, a 1.8% difference; Fig. 1), the price of the gentler envelope is paid almost entirely in time: traversal takes $\sim 44\%$ longer ($44.9 \rightarrow 64.5$ s). Formation quality is essentially retained—mean formation RMSE rises modestly from 0.176 to 0.217 m and the maximum is statistically indistinguishable (Fig. 3). The swarm does not hold its shape worse; it simply spends more time holding it. Fig. 4 summarizes the trade-off across all aggregate metrics: large, well-separated kinematic gains; a large time penalty; and formation/safety metrics that overlap within their error bars.

D. Qualitative Behavior

Fig. 5 shows the fish school mid-traversal and the full set of seven executed trajectories. The hexagon is held while the school weaves through the forest, with per-agent local occupancy clouds visible. Visually, the school glides rather than darts—consistent with the halved jerk—yet maintains the same spatial structure and clearance as the baseline.

VI. DISCUSSION

A. Interpretation

The experiment cleanly isolates a single design knob—the dynamic-feasibility envelope—and shows that it controls a graded, predictable trade-off: contracting it buys substantial smoothness and slower, lower-effort motion, and it costs time, with formation accuracy and safety essentially unchanged. For practitioners, this means a slow, drag-limited platform can reuse a high-performance quadrotor formation planner unchanged and obtain school-like motion simply by tightening $\{v_{\max}, a_{\max}\}$; the planner degrades gracefully rather than

TABLE II
AGGREGATE RESULTS ACROSS FIVE OBSTACLE SEEDS PER CONDITION (MEAN $\pm$ STD). ONLY THE DYNAMIC ENVELOPE DIFFERS BETWEEN COLUMNS.

Metric	Unit	Baseline (quadrotor)	Robotic fish	Change	Category
Max speed	m/s	1.581 ± 0.014	1.126 ± 0.047	−29%	Kinematic
Mean speed	m/s	0.976 ± 0.030	0.726 ± 0.068	−26%	Kinematic
Max accel	m/s ²	1.355 ± 0.249	0.854 ± 0.383	−37%	Kinematic
Mean accel	m/s ²	0.218 ± 0.025	0.123 ± 0.031	−44%	Kinematic
RMS jerk	m/s ³	0.658 ± 0.116	0.338 ± 0.129	−49%	Kinematic
Path length	m	51.420 ± 0.224	52.325 ± 0.372	+1.8%	Efficiency
Traversal time	s	44.919 ± 1.433	64.501 ± 4.327	+44%	Efficiency
Formation RMSE (mean)	m	0.176 ± 0.023	0.217 ± 0.026	+24%	Formation
Formation RMSE (max)	m	0.774 ± 0.078	0.799 ± 0.208	+3%	Formation
Min inter-agent dist	m	1.599 ± 0.354	1.705 ± 0.453	+7%	Safety
Mean obstacle clearance	m	1.116 ± 0.068	1.094 ± 0.077	−2%	Safety
Near-miss frac. (< 0.3 m)	–	0.058 ± 0.014	0.066 ± 0.013	+14%	Safety
Success rate	%	100	100	0	Safety

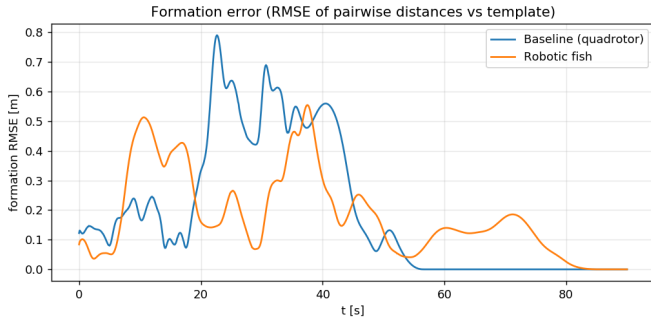


Fig. 3. Formation error (RMSE of pairwise inter-agent distances against the hexagon template) over time. The fish trace runs slightly higher but tracks the same envelope; the swarm holds shape in both conditions.

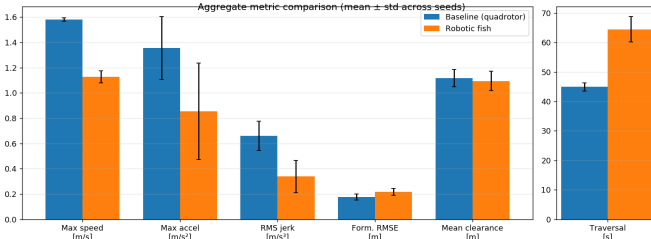


Fig. 4. Aggregate comparison (mean $\pm$ std over five seeds) for representative metrics. Kinematic smoothness improves substantially and traversal time grows; formation accuracy, clearance, and safety stay within noise.

failing or sacrificing formation/safety. The fact that formation RMSE barely moves while jerk halves indicates that the graph-similarity cost (2) is largely decoupled from the aggressiveness of the underlying motion—a useful robustness property.

B. Scope and Limitations

We are deliberately precise about what this study is and is not. (i) *Fidelity*. The agents use the second-order SO(3) quadrotor model with reduced limits, not a hydrodynamic, non-holonomic swimmer; the measured effects therefore reflect the dynamic *envelope*, not body-caudal-fin hydrodynamics, added mass, buoyancy, or ambient currents. The fish embodiment is a render-level re-skin. (ii) *Sample size*. Five seeds

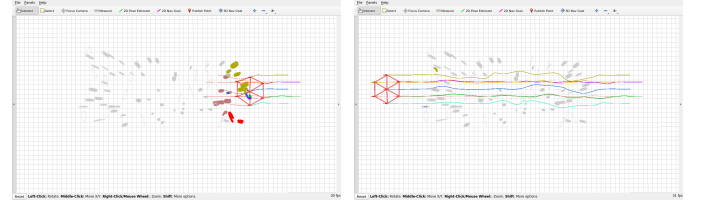


Fig. 5. Qualitative views (fish condition, seed 1). Left: the seven-agent hexagon school mid-field among the obstacle forest. Right: all seven executed trajectories spanning the field from the start hexagon to the goal.

per condition suffice to separate the large kinematic effects from their spread but render the near-equal formation/safety metrics “within noise,” not proven equal; a larger sweep with hypothesis tests would be needed to claim equivalence. (iii) *Single environment family*. All runs use one forest density and one start/goal geometry.

C. Toward Higher Fidelity

A natural extension replaces the SO(3) simulator and controller with a forward-swim model: a non-holonomic constraint (bounded turn rate, motion predominantly along the body axis), hydrodynamic drag and added-mass terms, buoyancy, and body-caudal-fin thrust [9], with an additional non-holonomic penalty term in the trajectory optimizer. Coupling this with implicit, vision-only coordination as demonstrated on physical fish swarms [7] would move the study from “school-like motion under a contracted envelope” toward true robotic-fish schooling, while the controlled-comparison methodology used here would carry over directly.

VII. CONCLUSION

We re-instantiated a graph-theoretic, seven-agent swarm formation planner as a school of robotic fish by changing only the visual embodiment and the dynamic-feasibility envelope, and quantified the consequences in a controlled, paired simulation study over five obstacle seeds per condition. Contracting the envelope to fish-like values roughly halves RMS jerk and cuts mean acceleration by $\sim 44\%$, yielding visibly smoother, lower-effort, school-like motion, at a cost of $\sim 44\%$ longer traversal

time and with formation accuracy, inter-agent spacing, obstacle clearance, and a 100% success rate all preserved within run-to-run noise. The dynamic envelope is thus a single, interpretable knob that trades time for smoothness without compromising formation or safety, and the planner degrades gracefully under a strongly bio-inspired regime. Future work will raise the dynamics to a true non-holonomic, hydrodynamic swimmer and broaden the environment sweep.

REPRODUCIBILITY

The planner modifications, headless experiment harness (data collection, metric computation, and figure generation), raw per-run logs, and all figures in this paper are available in the project repository.

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